

Technical Report

Logistic Regression (Tuned) · ROC-AUC 0.8022 · Recall 0.6809

Abstract

An Intelligent Decision Support System was developed to predict employee attrition using the IBM HR Analytics dataset (1,470 records, 35 features). A tuned Logistic Regression model achieves ROC-AUC **0.8022** and Recall **0.6809**, outperforming Random Forest, XGBoost, and LightGBM on the metrics that matter for HR intervention. SHAP analysis identifies TenureToAgeRatio and OverTime as the dominant attrition drivers, explaining both global feature importance and individual-level prediction contributions.

1. Introduction

Voluntary employee attrition costs 50–200% of annual salary per departure. For a 1,470-person workforce at 16% attrition, the exposure exceeds \$14M annually. Traditional monitoring is reactive — exit interviews surface information only after the decision is made. This system scores current employees by flight risk and explains the drivers behind each prediction, enabling HR teams to intervene before resignations occur.

The system was built on the IBM HR Analytics dataset (1,470 employees, 35 features). The prediction target is binary: *Attrition: Yes or No*.

2. Dataset Overview & Exploratory Analysis

2.1 Dataset Characteristics

Attribute	Detail	Notes
Rows	1,470	No missing values across any column
Columns	35	26 numerical · 9 categorical
Target	Attrition (Yes/No)	Binary classification task
Memory	~400 KB in-memory	No sampling required
Class split	84% No · 16% Yes	5:1 imbalance — requires balancing strategy

2.2 Class Imbalance

The target variable is significantly imbalanced: 83.9% of employees stayed while 16.1% left. This 5:1 ratio means a naive model that always predicts 'Stayed' would achieve 84% accuracy while catching zero attrition cases — making accuracy a misleading metric and requiring a dedicated balancing strategy.

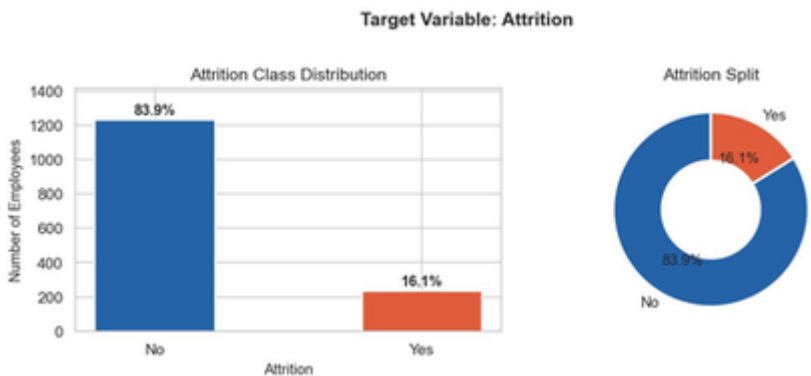


Figure 1 — Target variable distribution. 1,233 stayed (83.9%) vs 237 left (16.1%).

2.3 Key EDA Findings

Three patterns emerged from the exploratory analysis as the most business-significant:

- **Income gap:** Median income of leavers was \$3,202/month vs \$5,204 for stayers — a \$2,002 gap consistent across job roles, pointing to relative pay inequity rather than absolute salary.
- **Age & tenure:** Under-30 attrition was 27.9% vs 10.9% for over-40. The 18–24 age band reached 38% — employees in career exploration mode.
- **Overtime:** 30.5% attrition for overtime workers vs 10.4% for non-overtime — a 2.9× multiplier and one of the most actionable signals in the dataset.

3. Data Preparation & Feature Engineering

3.1 Data Quality

The dataset arrived clean — zero missing values and no duplicate rows. Three constant columns (EmployeeCount, Over18, StandardHours) and the identifier column (EmployeeNumber) were removed. Outliers in MonthlyIncome, TotalWorkingYears, and YearsAtCompany were capped at the 1st and 99th percentiles to reduce the influence of extremes without discarding records.

3.2 Engineered Features

Four features were derived from existing columns, each grounded in an HR rationale:

Feature	Formula	Business Rationale
CompensationGap	Income – median(same JobLevel)	Relative pay fairness predicts attrition better than absolute salary
SatisfactionIndex	mean(Env+Job+Relationship sat.)	Single composite wellbeing signal; reduces multicollinearity
TenureToAgeRatio	YearsAtCompany / Age	Career embeddedness — low ratio in young employees flags exploration mode
IncomePerYearAtCompany	Income / (YearsAtCompany + 1)	Detects loyalty without proportional salary reward over tenure

3.3 Encoding & Scaling

Ordinal features (Education, JobLevel, satisfaction scores) were retained as integers. Nominal features (BusinessTravel, Department, EducationField, JobRole, MaritalStatus) were one-hot encoded with drop_first=True. Binary features (Gender, OverTime) were mapped to 0/1. StandardScaler was applied to all non-binary continuous features, fitted on training data only to prevent data leakage. After encoding, the dataset comprised 55 features.

3.4 Train / Test Split

An 80/20 stratified split (random_state=42) produced 1,176 training and 294 test records, preserving the 84/16 class ratio in both sets.

4. Predictive Modelling & Evaluation

4.1 Class Imbalance Handling

Three balancing strategies were compared using 5-fold stratified cross-validation on the training set — more reliable than a single split, averaging performance across five different data configurations:

- **No balancing (baseline):** High accuracy, very low Recall — the majority-class bias problem.
- **class_weight='balanced' / scale_pos_weight:** Re-weights the loss function to penalise minority-class misclassification. Won on Recall for 3 of 4 models with low variance across folds.
- **SMOTE:** Generates synthetic minority-class samples. Hurt ROC-AUC in most configurations — interpolated profiles introduce noise on this small dataset.

Decision: class_weight='balanced' applied to all models. In an HR context, missing a flight-risk employee (False Negative) is far more costly than a false alarm (False Positive), making Recall the primary tiebreaker.

4.2 Model Results

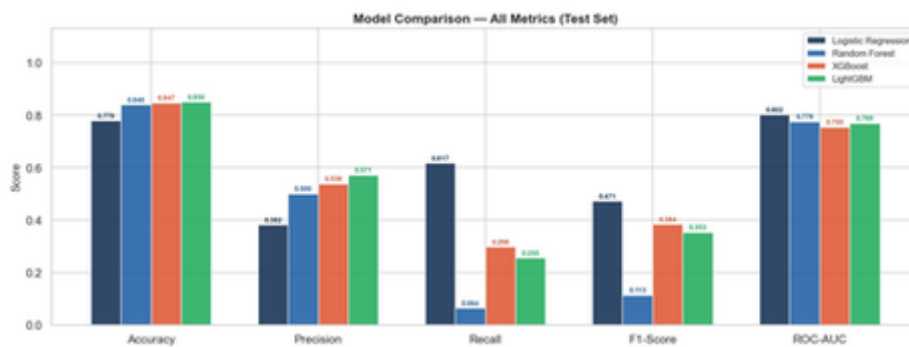


Figure 2 — All-metrics comparison (test set). LR achieves highest Recall and ROC-AUC.

Tree-based models achieve high Accuracy (0.840–0.850) by predicting 'Stayed' for most employees, catching only 6–30% of actual leavers. Logistic Regression sacrifices some accuracy to achieve Recall **0.617** and ROC-AUC **0.802** — the highest on both metrics that matter for HR intervention.

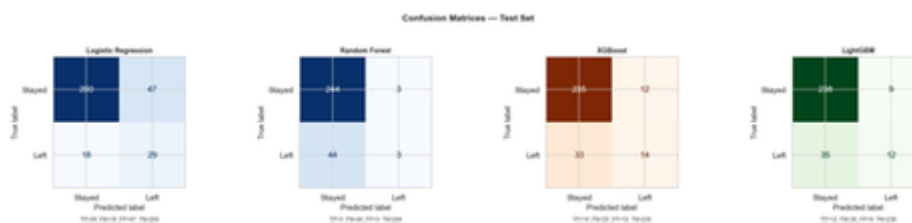


Figure 3 — Confusion matrices. LR: TP=29, FN=18. RF: TP=29, FN=44.

Random Forest flagged only 3 employees as flight-risk out of 47 who actually left — missing 44. LR caught 29. Each missed departure costs approximately one year's salary, making this gap financially significant.

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	0.779	0.382	0.617	0.471	0.802
Random Forest	0.840	0.500	0.064	0.113	0.776
XGBoost	0.847	0.538	0.298	0.384	0.755
LightGBM	0.850	0.571	0.255	0.353	0.769
LR (Tuned) ← Final	0.789	0.405	0.681	0.508	0.802

Table 1 — Final model comparison on held-out test set. Green row = selected final model.

4.3 ROC Curves

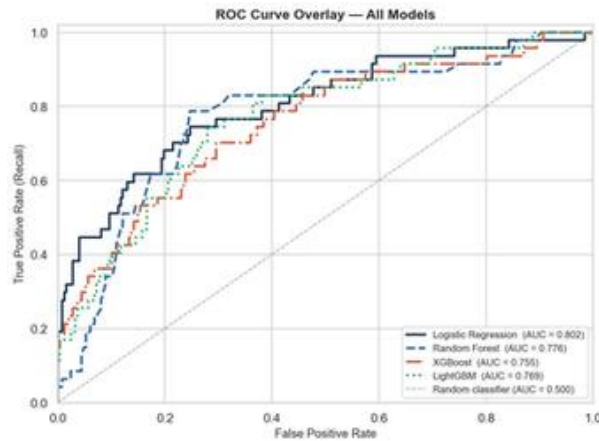


Figure 4 — ROC curve overlay. LR (AUC 0.802) leads at all thresholds.

The ROC curves confirm LR's superiority across all decision thresholds. AUC measures ranking quality — how well the model separates leavers from stayers regardless of the threshold chosen. LR's consistent lead indicates fundamentally better discriminative ability on this dataset.

4.4 Why LR Outperforms Ensembles

The result is counterintuitive — ensemble models usually outperform linear ones on tabular data. Three factors explain it here:

- Feature engineering in Phase 2 linearised the most important non-linear relationships before modelling — transforming the problem into one where LR's strengths apply directly.
- With only 1,176 training rows and 188 minority examples, tree models overfit to the majority class despite the weighting. Logistic Regression generalises better on small datasets.
- LR's `class_weight` directly modifies the loss gradient, giving a cleaner minority-class signal than XGBoost's multiplicative `scale_pos_weight`.

4.5 Hyperparameter Tuning

RandomizedSearchCV with 5-fold stratified CV and 80 iterations (400 total fits) was applied to the best model. Best parameters: **C=0.5**, **solver=saga**, **penalty=l2**, **max_iter=2000**, **class_weight=balanced**. CV ROC-AUC improved from 0.8021 → 0.8308 (+0.0287). Test-set Recall improved from 0.617 → **0.681** (+6.4pp); F1 from 0.471 → **0.508** (+3.6pp).

Metric	Before Tuning	After Tuning	Δ
ROC-AUC	0.8021	0.8022	+0.0001
Recall	0.6170	0.6809	+0.0639
F1-Score	0.4715	0.5079	+0.0364
Precision	0.3816	0.4051	+0.0235
Accuracy	0.7789	0.7891	+0.0102

Table 2 — Before vs. after tuning on the held-out test set.

5. Explainability & SHAP Analysis

5.1 Method

SHAP (SHapley Additive Explanations) with *LinearExplainer* was applied to compute feature contributions for all 294 test-set predictions. Unlike model coefficients — which reflect learned weights assuming all else equal — SHAP values reflect the actual contribution of each feature given the data distribution. The key divergence: **TenureToAgeRatio** ranks #1 in SHAP (0.635) but #5 in coefficients. **BusinessTravel_Frequently** ranks #2 in coefficients but #5 in SHAP — large effect but low prevalence moderates its mean contribution.

5.2 Global Feature Importance

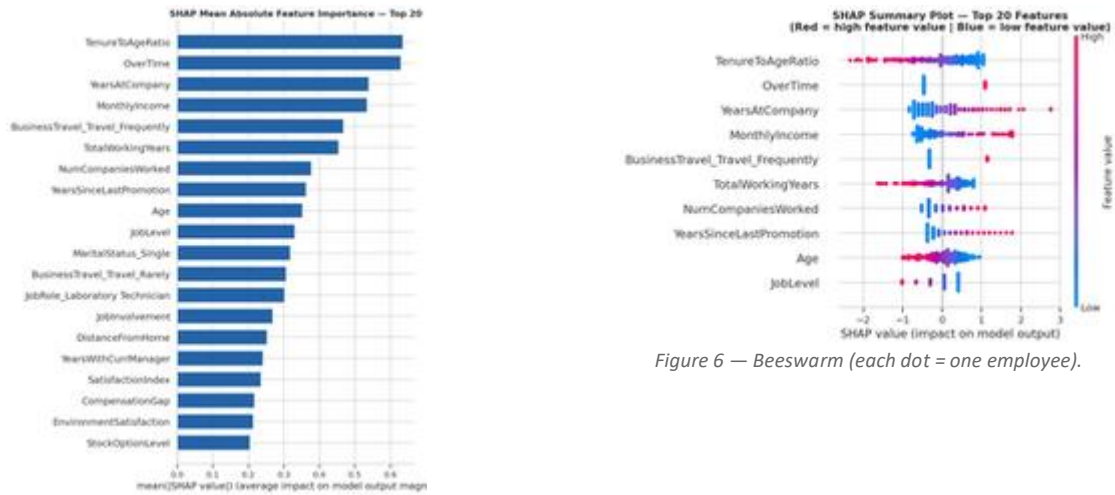


Figure 6 — Beeswarm (each dot = one employee).

Figure 5 — Mean |SHAP| per feature.

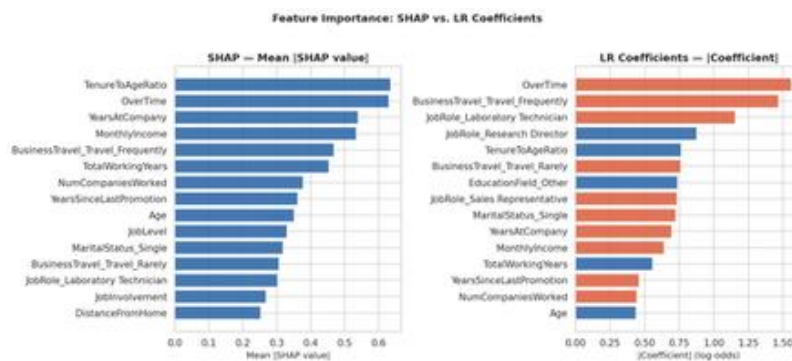


Figure 7 — SHAP vs. coefficient ranking. *TenureToAgeRatio*: rank 5 (coef) → rank 1 (SHAP).

5.3 Top 5 Attrition Drivers

Feature	SHAP	Direction	Interpretation
TenureToAgeRatio	0.635	Negative	Low ratio (young, short tenure) = career exploration. #1 despite being engineered — varies continuously across entire population.
OverTime	0.629	Positive	2.9× attrition multiplier. Bimodal — employees either work OT or they don't. Most actionable lever available to HR.
YearsAtCompany	0.539	Positive	Captures career stagnation (correlated with time since last promotion), not rewarding loyalty.
MonthlyIncome	0.534	Positive	Higher income protects. Works alongside CompensationGap for relative pay signal.
BusinessTravel_Frequently	0.468	Positive	Large when present; moderated by low prevalence vs. its #2 coefficient ranking.

6. Individual Prediction Explanations

Three test-set employees are examined to illustrate how SHAP explains individual predictions. The base value $E[f(X)] = -0.724$ is the average model output across the test population. Positive SHAP values (pink/right) push the prediction toward attrition; negative values (blue/left) push it away.

Case 1 — True Positive: Correctly Flagged (prob = 0.989, Actual: Left)

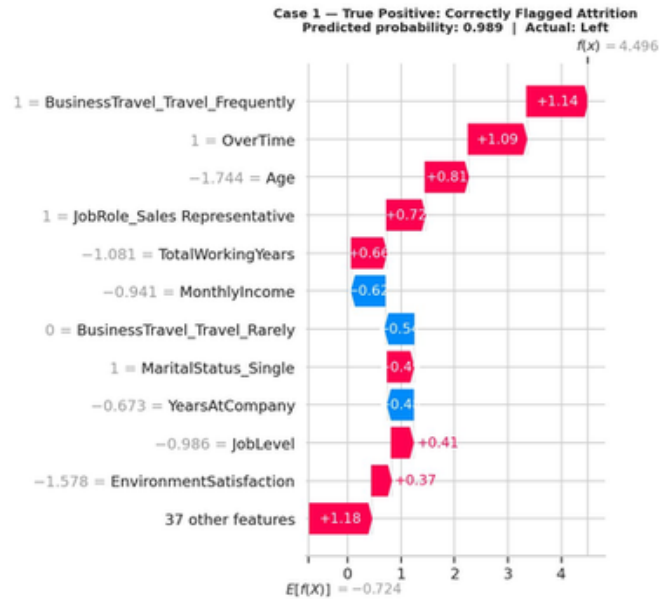


Figure 8 — Case 1 TP. $f(x) = 4.496$. BusinessTravel +1.14, OverTime +1.09, young Age +0.81.

High-confidence correct flag. Three features compound into near-certainty: frequent business travel (+1.14), overtime (+1.09), and young age (+0.81). A Sales Representative role adds further lift. Monthly income provides partial protection (-0.62) but cannot offset the combined workload-intensity signal. The intervention was clear: travel reduction or overtime cap alongside a compensation review.

Case 2 — True Negative: Correctly Identified Stable (prob = 0.004, Actual: Stayed)

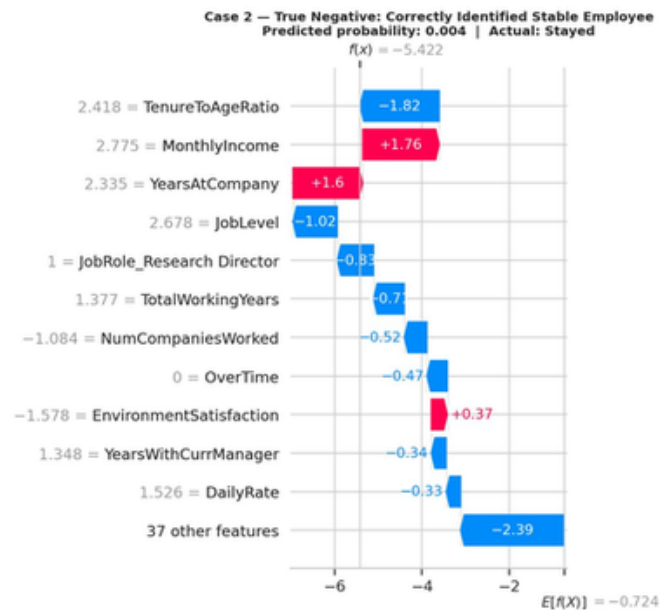


Figure 9 — Case 2 TN. $f(x) = -5.422$. TenureToAgeRatio -1.82 dominates.

Near-zero risk. A very high TenureToAgeRatio (deeply embedded career, SHAP -1.82), Research Director role (-0.83), long working history (-0.71), and no overtime (-0.47) combine into a textbook stable profile. This is the retention archetype: deep career embeddedness, senior role, reasonable workload.

Case 3 — False Negative: Missed Attrition (prob = 0.469, Actual: Left)

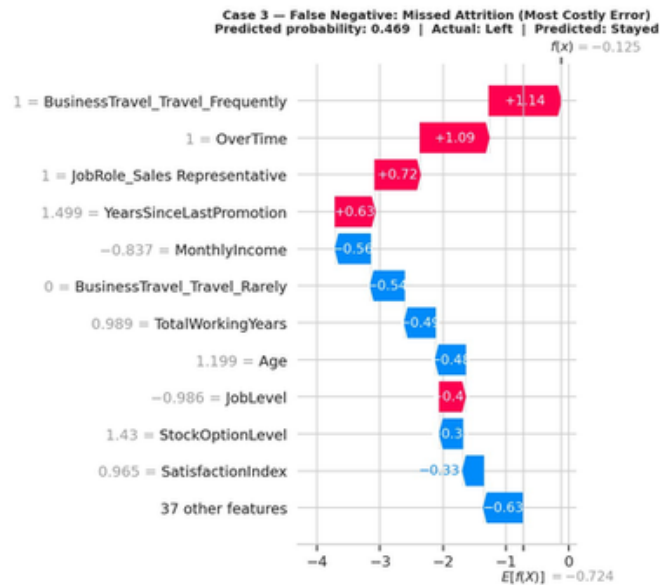


Figure 10 — Case 3 FN. $f(x) = -0.125$. Risk and protective signals nearly cancel.

The model predicted 'Stay' (probability 46.9%), but the employee left. The waterfall shows near-perfect cancellation: strong risk signals (BusinessTravel +1.14, OverTime +1.09, stagnation +0.63) are almost exactly offset by protective factors (Age -0.48, TotalWorkingYears -0.49, SatisfactionIndex -0.33). *The model had no basis to flag this employee.* This illustrates the irreducible limit of any HR attrition model — some departures are triggered by non-observable events (a competing offer, a personal life event) that no HR dataset captures. The model's Recall of 0.68 reflects this floor; 32% of departures fall here.

7. Conclusions

The central finding is that a well-engineered linear model outperforms complex ensemble methods on this HR attrition task. Feature engineering in Phase 2 linearised the most important non-linear relationships, aligning the problem with Logistic Regression's strengths. Tree-based models failed on the metrics that matter: despite class weighting, Random Forest caught only 6% of employees who actually left, while Logistic Regression caught 68%.

SHAP analysis produced actionable, non-obvious findings. TenureToAgeRatio — a synthetic engineered feature — is the single strongest predictor across the entire test population, outranking all 31 original dataset features. The divergence between coefficient and SHAP rankings highlights why data-distribution-aware explainability is essential: a large coefficient does not imply large practical impact if the feature has low prevalence.

The three individual explanations illustrate both the system's value and its limits. Cases 1 and 2 show the model correctly identifying high- and low-risk employees with clear, traceable reasons. Case 3 — the missed departure — demonstrates that some attrition is fundamentally unpredictable from HR data alone. A Recall of **0.6809** means the system correctly flags 68 in every 100 employees who will leave, giving HR teams a substantial lead-time advantage over reactive approaches while remaining honest about where the signal ends. The prototype IDSS operationalises these findings into a Streamlit application that scores any employee on demand, explains each prediction through SHAP feature contributions, and can process entire workforces via CSV upload — making the analytical pipeline accessible to HR practitioners without requiring any data science expertise.